

devices like gaming consoles, phones and tablets on the network. NAS systems can also be configured as FTP, Web, e-mail, and print servers.

Hence, the decision of choosing the right NAS solution becomes even more complicated. Here's all you need to know while selecting the right network-attached storage.

Capacity and expandability

NAS systems are typically for home users and small to mid-sized businesses. While those for home users are available with a storage capacity of almost 8TB, for businesses, the NAS boxes scale higher and go up to even petabyte levels. So, for a business, one should first judge not only the present storage needs but also keep in mind the future requirements. This is where the expandability feature of a NAS system comes in. Bhattacharya asserts, "NAS systems available in the market are as small as 2TB or 4TB and as high as 360TB. The buyer must have a very clear understanding of how much storage will be needed. Often, what we've seen is that people have a problem with correctly estimating their storage needs and then the problem of expandability comes in. The IT decision-makers should have an idea of their storage needs and then buy an appropriately expandable NAS solution."

Amongst the expandability options, users get USB ports and iSCSI support, which can be used for attaching direct storage and creating virtual drives, respectively.

Network connectivity

Most NAS solutions offer wired gigabit Ethernet connectivity. The enterprise grade NAS products have two or more gigabit ports, which can be used for port trunking. This helps in providing connection redundancy if one port fails. Moreover, if your NAS solution has multiple ports, they can be link-aggregated, leading to an increase in network throughput by combining the link speed of the ports.

In addition, some companies also offer wirelessly accessible NAS solutions. Though, wireless NAS solutions experience the latency issue more often than the wired ones, when it comes to transferring large files.

Backup and recovery of data

Bhattacharya explains, "NAS is for recovering data when something goes wrong. The most important feature that buyers should look at is whether or not the system has a good snapshot facility. For example, let's look at a scenario in which there are 10 laptops in a network and there is a central storage NAS, which keeps the backup of all the laptops connected with it. The NAS solution takes periodic snapshots. When it does this, it means that for that moment and snapshot, whatever is stored in the laptop is also replicated in the NAS. So, if a user loses his data from the

laptop at any point in time, he can replicate the entire data to the laptop by simply going to the last snapshot. At the most, the user may lose only the work done after the snapshot was taken. So, the more snapshots that are taken, the more accurate are the store points."

Performance

According to Khwaja Saifuddin, senior director - South Asia, Middle East and Africa, Western Digital (WD), opines, "Different businesses have different storage requirements. However, a few needs such as affordability, scalability, reliability and performance remain common." As in the case with PCs, the performance of NAS units too depends upon processors and memory. The better the processor and the more the installed memory, the better a NAS solution will perform. But, as they say, quality comes at a price. For higher specifications, you may have to pay a higher price as well.

If the users in a company require a lot of I/O operations, it is good to go for a NAS solution with a high-end processor and more memory.

Security

The file system that the NAS system uses should support a high degree of data integrity. One of the biggest problems of any storage system is data loss caused by unstable file systems. Over a period of time, companies end up storing a lot of data in the NAS. So the file system must have an operating system with more data integrity like ZFS or BTRFS, which are comparatively more stable than others.

Warranty

Rajesh Khurana, country manager, Seagate, shares, "The warranty of a NAS solution is one important aspect that an IT decision maker should look at. One should buy products from a stable company that offers a good warranty, because this demonstrates the confidence of the manufacturer."

Remote access

NAS devices are no more restricted to local storage or access needs. Keeping the trend of remote access and storage in mind, modern day NAS solutions come with remote access capabilities that help in managing the device and enable remote access to data. They also offer cloud services, which are useful for sharing content, irrespective of the location.

Power consumption

Power consumption becomes an important factor in the long run. Saifuddin shares, "There are NAS solutions designed with innovative technologies that reduce power consumption and lower the operating temperature. This significantly improves the overall drive performance and reliability."